

NATIONAL JUNIOR COLLEGE, SINGAPORE
Senior High 2
Preliminary Examination
Higher 2

CANDIDATE
NAME

BIOLOGY
CLASS

REGISTRATION
NUMBER

Biology

9744/02

Paper 2 Structured Questions

23 August 2018

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your name, Biology class and registration number on all the work you hand in.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in the brackets [] at the end of each question or part of question.

For Examiner's Use	
Section A	
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This document consists of **31** printed pages.

Section A

Answer the question in this section.

- 1 Fig. 1.1 shows a section of a cell surface membrane.

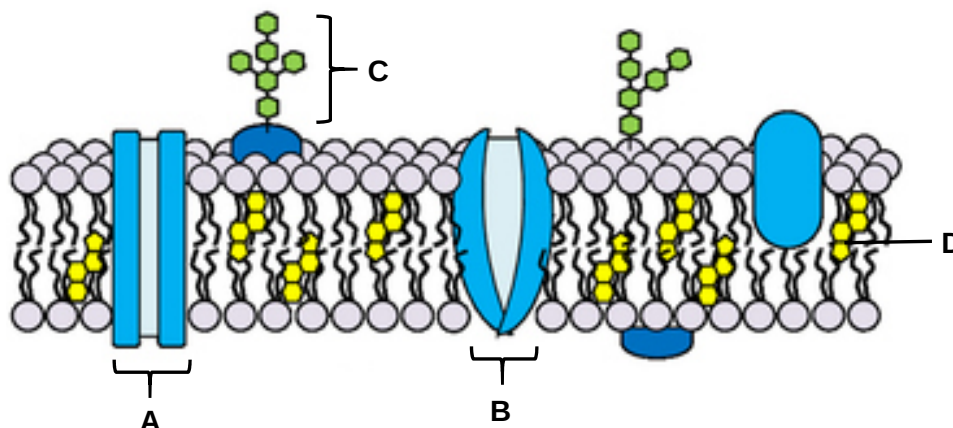


Fig. 1.1

- (a) Name the structures labelled A, B, C and D.

(4 correct – 2 marks; 2 or 3 correct – 1 mark; 0 or 1 correct – 0 mark)

A **channel protein**

(Reject: transmembrane / integral / intrinsic membrane protein)

B **carrier protein**

(Reject: transmembrane / integral / intrinsic membrane protein)

C **oligosaccharide / sugars / carbohydrate**

(Reject: glycoprotein)

D **cholesterol**

[2]

- (b) Describe how structures A and B are held in the membrane.

1. **hydrophobic interactions between the non-polar hydrocarbon tails of the phospholipid bilayer and the hydrophobic R groups of the non-polar amino acids in the exterior surface of the proteins;**
2. **hydrophilic interactions between the phosphate heads of phospholipids in the bilayer and the polar and charged R groups of amino acids in the exterior surface of the proteins;**

[2]

- (c) For hydrophilic molecules to enter a cell, they require the help of either structure A or B.

State and explain which of the two structures allows a faster entry into the cell.

1. Structure A (channel protein);
2. Hydrophilic molecules do not need to bind to the channel protein in order to enter the cell;
3. Channel protein does not need to undergo any conformational change to allow the entry of the hydrophilic molecules into the cell;
4. Carrier protein, on the other hand, requires the hydrophilic molecules to bind to it before it undergoes a conformational change that results in the transport of the hydrophilic molecules into the cell;

[3]

- (d) State two possible functions of structure C.

1. increases the hydrophilic characteristics of lipids and proteins;
2. stabilises the conformation of many membrane proteins;
3. contributes to cell-cell recognition / communication;
4. contributes to cell-cell adhesion;
5. contributes to signal transduction;
6. used as antigens in the body's immune responses;
7. protects the cell membrane from mechanical damage;
8. AVP

[2]

- (e) Suggest why there seems to be a greater diversity in the molecular structures of A and B (proteins) than that of C (carbohydrates).

1. greater variety of monomers – at least 20 different amino acids / variety due to side chains or R groups;
2. more types of bonds – hydrogen bonds, ionic bonds, disulphide bonds, hydrophobic interactions;
3. more levels of structure – primary, secondary, tertiary, quaternary;

[2]

- (f) The fluid mosaic model was first proposed by S.J. Singer and Garth L. Nicolson in 1972 to explain the structure of the cell surface membrane.

Explain why it is called fluid mosaic.

1. fluid – phospholipids and proteins free to move laterally along the membrane;
2. mosaic – proteins embedded / studded / scattered in the phospholipid bilayer OR phospholipids and proteins distributed asymmetrically across the bilayer;

[2]

- (g) Comment on the significance of structure D in the cell surface membrane.

1. maintains membrane fluidity when temperature changes OR provides mechanical stability;
2. prevents leakage of polar molecules;

[1]

[Total: 14]

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Section B

For Examiner's Use	
Section B	
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3	/12

Section B

Answer **all** the questions in this section.

- 2 In an experiment to investigate the effect of insulin on glucose uptake by muscle cells, the concentration of free glucose inside muscle cells was measured with respect to the extracellular glucose concentration, in the presence of insulin (labelled as insulin) and in the absence of insulin (labelled as control).

Fig. 2.1 shows the results of the experiment.

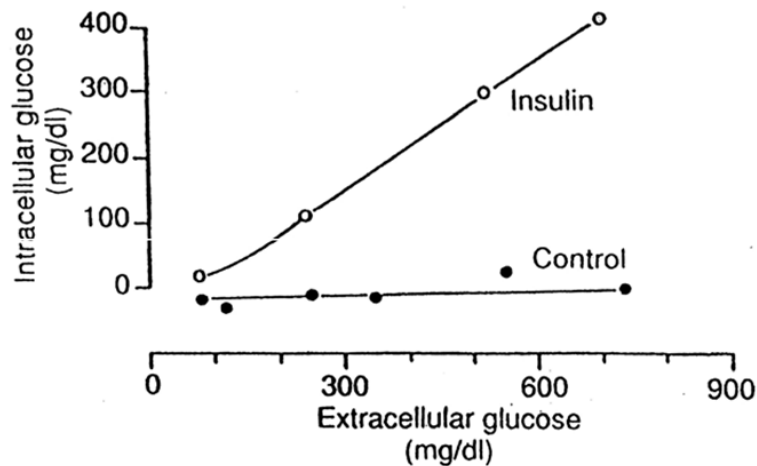


Fig. 2.1

- (a) With reference to Fig. 2.1, describe the effect of insulin on glucose uptake by muscle cells.

1. Insulin leads to an increase in glucose uptake / intracellular glucose concentration in the muscle cells;
2. In the absence of insulin, intracellular glucose concentration remains constant at about 0 mg/dl as extracellular glucose concentration increases from 78 (70 – 80) to 733 (720 – 740) mg/dl;
3. In the presence of insulin, intracellular glucose concentration increases from 22 (10 – 30) to about 400 (400 – 420) mg/dl as extracellular glucose concentration increases from 78 (70 – 80) to about 700 (690 – 710) mg/dl;

[3]

- (b) (i) Name the type of receptor that insulin binds to on the muscle cell.

Receptor tyrosine kinase / Tyrosine kinase receptor;
(Reject: Insulin receptor; RTK)

[1]

(ii) Explain how the binding of insulin to its receptor could result in the effect shown in Fig. 2.1.

1. The binding of insulin to its receptor would induce a conformational change in the receptor, bringing the two internal tyrosine kinase domains closer together;
(Reject: cause receptor dimerisation)
2. Contact between the two adjacent tails of the receptor would activate their tyrosine kinase function, leading to cross-phosphorylation / autophosphorylation of the tyrosine residues present in the tails of the receptor;
3. The fully activated receptor would trigger the assembly of adaptor proteins on the receptor tails, which will further recruit and activate other downstream relay molecules via phosphorylation;
4. More glucose transporters (GLUT4) would become embedded in the plasma membrane, increasing the glucose uptake / intracellular glucose concentration in the muscle cells;

[4]

(c) Suggest how the effect of insulin on glucose uptake by muscle cells could be terminated.

1. degradation of insulin by enzymes;
2. endocytosis of insulin-RTK / ligand-receptor complex;
3. increase in the activity of phosphatases / enzymes that dephosphorylate proteins;

[1]

[Total: 9]

- 3 Succinate dehydrogenase is an enzyme in the Krebs cycle, which catalyses the conversion of succinate to fumarate by dehydrogenation.

Malonate is an inhibitor of succinate dehydrogenase. An experiment was carried out to investigate the effect of malonate on respiration. Isolated liver mitochondria were placed in six reaction tubes, with contents as shown in Table 3.1. The corresponding rates of oxygen uptake were measured and also tabulated.

Table 3.1

tube	volume of substance added / cm ³				rate of oxygen uptake / arbitrary units
	buffered liver mitochondria suspension	2% glucose solution	2% pyruvate solution	0.2% malonate solution	
1	2.00	0.00	0.00	0.00	1.1
2	2.00	0.01	0.00	0.00	1.2
3	2.00	0.00	0.01	0.00	17.8
4	2.00	0.00	0.01	0.01	6.7
5	2.00	0.00	0.01	0.02	2.2
6	2.00	0.00	0.04	0.02	15.5

- (a) Explain why rate of oxygen uptake can be used as an indicator of rate of respiration.

Oxygen is the final electron acceptor in oxidative phosphorylation; [1]

- (b) Explain why glucose has no effect on the rate of oxygen uptake.

- Enzymes for glycolysis were absent in Tube 2;**
- Hence, glucose was not oxidised to pyruvate / no pyruvate was present for Krebs cycle to take place;** [2]

- (c) (i) With reference to Table 3.1, describe the effect of adding pyruvate to tube 3.

- Adding pyruvate increased the rate of oxygen uptake by about 16 folds (highest among the 6 tubes);**
- The rate of oxygen uptake increased from 1.1 arbitrary units in Tube 1 to 17.8 arbitrary units in Tube 3;** [2]

(ii) Explain your answer to (c) (i).

1. Pyruvate was converted to acetyl CoA in the link reaction;
2. Acetyl CoA then entered the Krebs cycle, which produced NADH and FADH₂;
3. NADH and FADH₂ then passed their electrons down the electron transport chain until they reached molecular oxygen, which was reduced to water;

[3]

(d) Account for the difference in rate of oxygen uptake between tubes 5 and 6.

1. In the presence of malonate, rate of oxygen uptake increased from 2.2 arbitrary units in tube 5 to 15.5 arbitrary units in tube 6 when the volume of pyruvate added increased from 0.01 cm³ in tube 5 to 0.04 cm³ in tube 6;
2. Malonate acted as a competitive inhibitor of succinate dehydrogenase / competed with succinate for the active site of succinate dehydrogenase;
3. The inhibitory effect of malonate could be overcome by increasing the concentration / volume of pyruvate, which consequently gave more succinate;
4. When substrate / succinate concentration increased, the frequency of effective collisions between substrate / succinate and enzyme / succinate dehydrogenase molecules was higher than that between substrate / succinate and inhibitor / malonate molecules;

[4]

[Total: 12]

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Section C

For Examiner's Use	
Section C	
4	/8
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Section C

Answer **all** the questions in this section.

- 4 (a) Outline the cell theory.

1. ref to smallest unit of life;
2. ref to all cells come from pre-existing cells;
3. ref to living organisms are composed of cells;

[3]

- (b) Endosymbiotic theory holds that the organelles distinguishing eukaryotic cells evolved through symbiosis of unicellular prokaryotic cells.

Fig. 4.1 shows the electron micrograph of a photosynthetic bacterium, *Rhodospseudomonas viridis* (x60,000). The internal photosynthetic membranes extend the entire length of the bacterium and are visible in cross section on the right of the cell, and in tangential section on the left of the cell.



Fig. 4.1

- (i) Calculate the actual length of the bacterium shown in Fig. 4.1.

Draws a line on bacteria cell to indicate length

Length = magnified size/60000 =

[1]

- (ii) With reference to the visible structures in Fig. 4.1, describe how the structure of *R. viridis* supports and does not support the endosymbiotic theory.

Support:

Stacked membrane structure resembles thylakoid stacks in a chloroplast;

Multiple stacks within cell resembles the multiple thylakoid stacks in a chloroplast;

[2]

Does not support:

presence of cell wall, which is not found in a chloroplast;

- (iii) Describe **two** other lines of evidence that support the endosymbiotic theory, but may not be visible in Fig. 4.1.

- 1. 70S ribosomes found in bacteria and chloroplast;**
- 2. Circular DNA found in bacteria and chloroplast;**

[2]

[Total: 8]

- 5 Bacteriophages play an important role in regulating microbial ecology of many ecosystems because of their impact on bacteria.

Fig. 5.1 shows the electron micrograph of some bacteriophages.

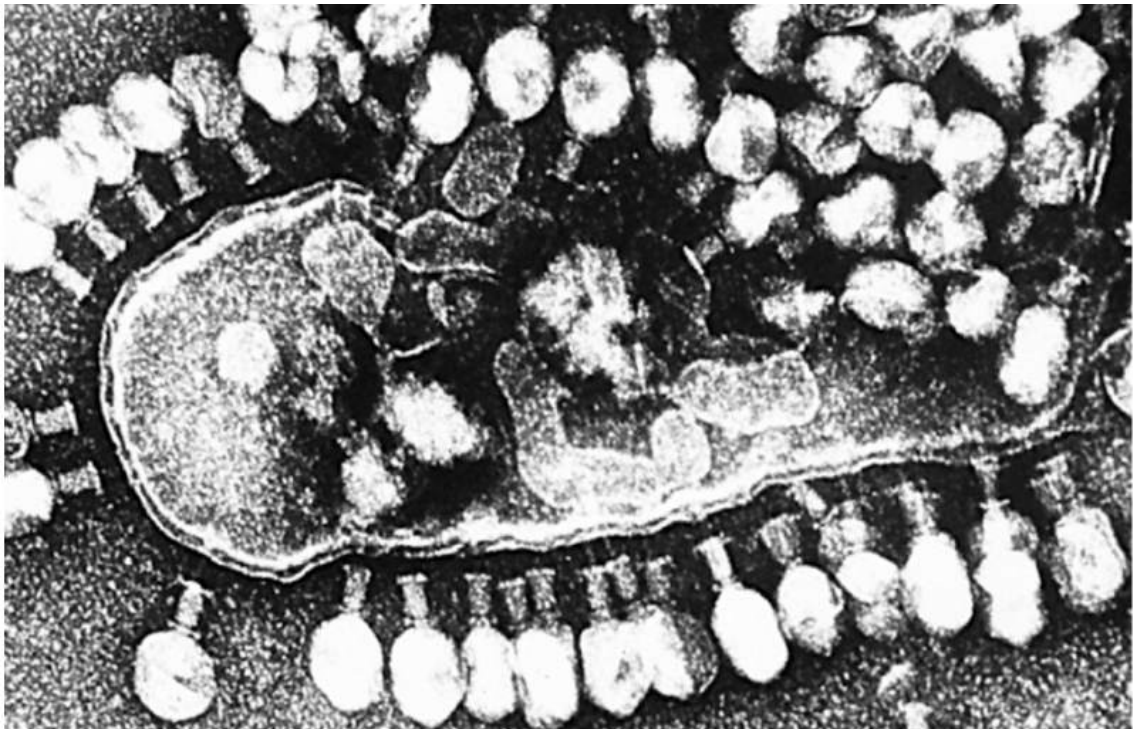


Fig. 5.1

- (a) (i) Identify the reproductive cycle shown in Fig. 5.1.

Lytic cycle;

[1]

- (ii) Describe how the bacteriophage genome is inherited in the reproductive cycle identified in (a)(i).

1. Tail sheath contracts and injects phage genome into host cell;
2. Host cell machinery directed to synthesize phage protein and to replicate phage genome;
3. Phage proteins assembled to form phage heads, tails and tail fibres;
4. Phage genome is packaged inside capsid during assembly of phage heads;
5. Phage lytic enzyme damages peptidoglycan cell wall + osmotic lysis + release of phage particles;

[3]

(b) Describe the process through which bacteriophages may contribute to bacterial survival.

1. Gene transfer between bacterial cells by transduction;
2. Packaging of bacterial DNA inside capsid during assembly of new viral particle;
3. Donor bacterial DNA may carry advantageous alleles for new bacterial host cell;
4. Recombination between donor DNA and host DNA may result in new advantageous phenotype;

[3]

(c) Describe **two** similarities between the reproductive cycles of bacteriophages and animal viruses.

1. Specificity for host cell + ref to binding to specific host receptors;
2. Directs synthesis of new viral particles using host cell machinery;

[2]

[Total: 9]

- 6 The term “operon” was coined by Jacob and Monod, who characterised the first defined classical operon, the *lac* operon, in *Escherichia coli*. While working to elucidate the *E. coli lac* operon, Monod and his colleagues developed a range of biochemical tools.

(a) Outline the features of an operon and explain how gene expression may be regulated in general.

1. Promoter + operator + structural genes + terminator;
2. Ref to structural genes involved in related biochemical pathway;
3. Ref to inducible operon + transcription turned on by effector molecule;
4. Ref to repressible operon + transcription turned off by effector molecule;
5. Ref to effector molecules, such as activators/repressors + bind to DNA sequence + turn on/off transcription;

[3]

(b) Isopropyl- β -D-thiogalactoside (IPTG) was used in Monod's experiments to identify the *lac* repressor protein, and to induce the *lac* operon.

Fig. 6.1 shows the molecular structure of lactose and IPTG.

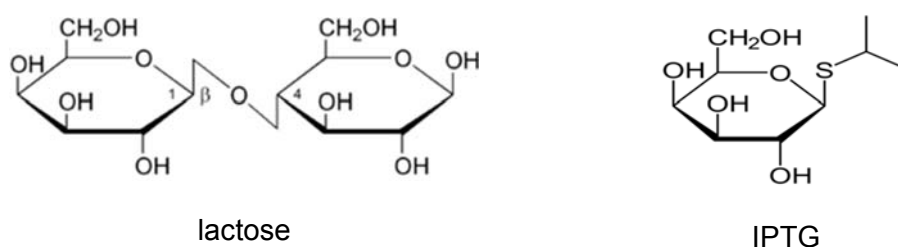


Fig. 6.1

With reference to Fig. 6.1, explain how IPTG is able to induce the *lac* operon and suggest **one** advantage of using IPTG instead of lactose in Monod's experiments.

1. ref to structural similarity between IPTG and galactose;
2. β -galactosidase cannot cleave IPTG, hence inducer concentration will remain constant through experiment;

[2]

- (c) The discovery of F factors carrying the *lac* operon allowed Monod to construct stable partial diploids for his experiments.

Four strains of bacteria were grown in glycerol (no glucose), with or without IPTG.

Table 6.1 shows the results of some of Monod's experiments. The plus sign or minus sign indicates presence or absence of β -galactosidase or permease activity.

Table 6.1

strain	genotype	β -galactosidase		permease	
		without IPTG	with IPTG	without IPTG	with IPTG
1	$O^+Z^+Y^+$	-	+	-	+
2	$O^+Z^+Y^+ / F(O^+Z^-Y^+)$	-	+	-	+
3	$O^{m1}Z^+Y^+$	+	+	+	+
4	$O^+Z^-Y^+ / F(O^{m1}Z^+Y^-)$	+	+	-	+

O^+ represents wildtype operator

O^{m1} represents mutant operator

Z^+ represents wildtype *lacZ*

Z^- represents no *lacZ*

- (i) From the results of experiments using strains 1 and 2, Monod concluded that the wildtype is inducible and Z^+ is dominant to Z^- .

Complete Table 6.1 to show the results that will support the above conclusions.

[1]

- (ii) State and explain **one** conclusion that could be drawn from the results of experiments using strains **3** and **4**.

1. **O^{m1} is a constitutive mutation (OWTTE);**
2. **ref to polymerase always bound to operator and allowing expression of the structural genes;**

OR

3. **operator is cis-acting (OWTTE);**
4. **Y is only expressed when induced but Z is induced with or without IPTG;**

[2]

- (iii) Suggest and explain how the results of experiments using strain **1** would differ if the bacteria were grown in glycerol and glucose.

1. **ref to results -/-/-/- with and without IPTG;**

AND

2. **ref to low cAMP in presence of glucose + inactive CAP;**
3. **ref to inactive CAP + not binding to CAP-binding site on promoter + no transcription;**

[2]

[Total: 10]

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Section D

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Section D

Answer **all** the questions in this section.

- 7 (a) Define and state an example of a multipotent stem cell.

1. Differentiate into cell types of a tissue / organ;
2. Haematopoietic stem cell;

[2]

- (b) (i) One unique feature of stem cells is their ability to differentiate into different cell types, where some genes are silenced and some genes are expressed.

Explain how stem cells achieve this.

1. DNA methylation occurs on the CG islands of DNA;
2. Recruits Histone Deacetylases;
3. Histone deacetylation on lysine residues of histone tails;
4. Increases the positive charge of histone tails and increases electrostatic forces of attraction between histone tails and DNA;
5. Decrease gene expression / silencing of certain genes resulting in differentiation;

[Opposite way of expressing idea will be awarded equivalent marks]

[4]

- (ii) During the process of differentiation, specific proteins are produced.

Account for the roles of **three** types of RNA in producing these proteins.

1. mRNA – carries genetic information from DNA in the form of a series of codon to specify the amino acid sequences of proteins;
2. tRNA – serves as an adaptor molecule in protein synthesis; carries the correct amino acid to the ribosome and its anti-codon base-pairs with its complementary mRNA codon;
3. rRNA – component of ribosome, plays structural and catalytic role in ribosomes;

[3]

- (c) A recent advancement in stem cell gene therapy is the use of CRISPR/Cas9 system to edit genes. This stem cell gene therapy can help to cure genetic diseases by removing the undesired gene and adding the corrected version in the stem cells.

The CRISPR/Cas9 system works by delivering a Cas9 nuclease complexed with a single-stranded RNA (artificial guide) into a cell.

Fig. 7.1 and Fig. 7.2 explain how gene editing is achieved using CRISPR/Cas9 system.

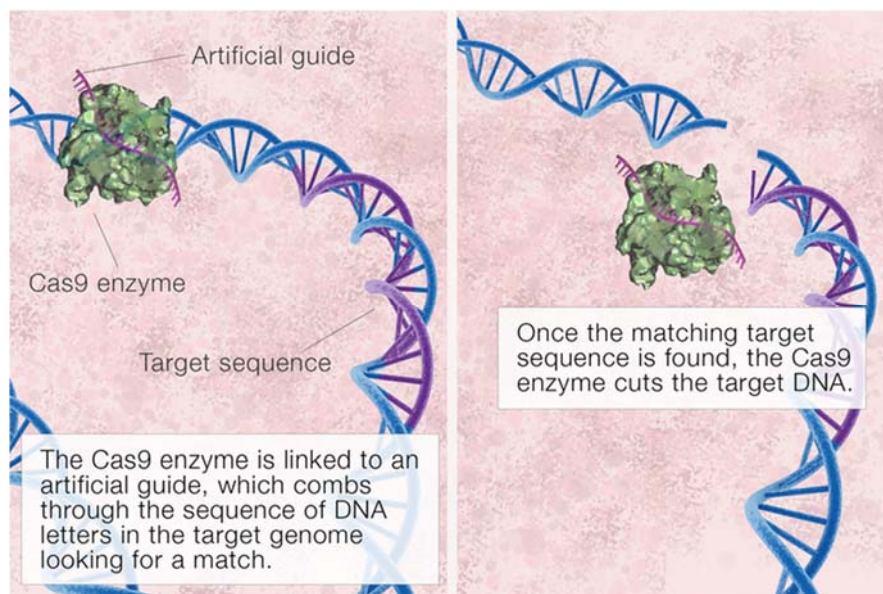


Fig. 7.1

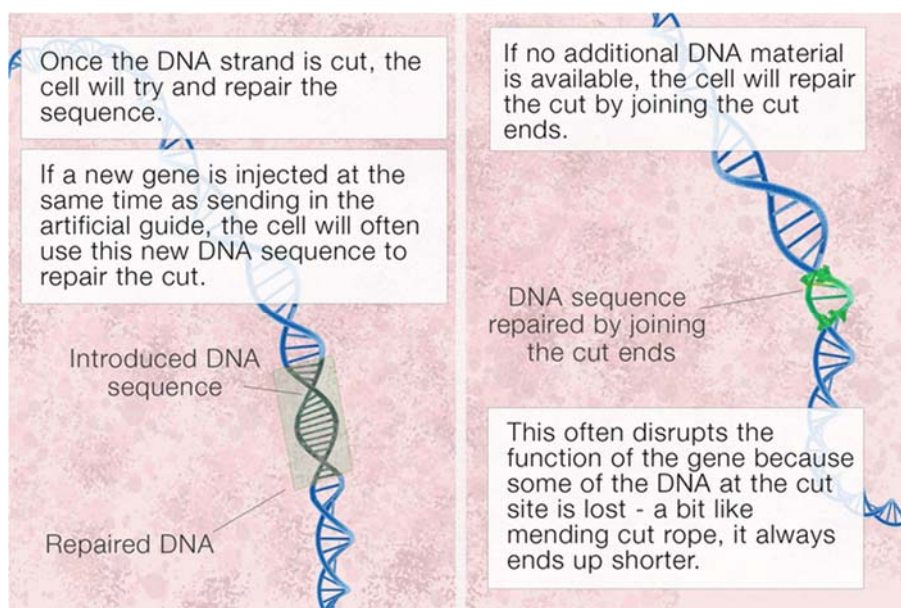


Fig. 7.2

- (i) With reference to Fig. 7.1 and Fig. 7.2, explain why CRISPR/Cas9 is believed to have potential applications for treating **many** genetic diseases in humans.

1. Different genetic diseases are caused by different genes;
2. Use of artificial guide allows Casp9 enzyme to find the right gene based on complementary base-pairing of the guide to the target gene to edit the gene;

[2]

- (ii) Despite the potential of CRISPR/Cas9 in treating many genetic diseases, some scientists are worried about the use of such technology in humans.

Suggest **one** possible consideration.

1. Possible off-target mutations (unintended mutations) in the genome. Mutations are deleterious;
2. Cost of germline editing technology is very high to the extent that only families from rich countries could afford it;
3. Genome editing in human embryos could have unpredictable effects to the future generation;
4. Technology could be used for non-therapeutic modifications, leading to loss of human diversity and eugenics;
5. AVP

[1]

[Total: 12]

- 8 (a) A single coral species is often spread across heterogeneous environments, and populations that experience different temperature regimes can have markedly different responses and thresholds to thermal stress.

A study was conducted to investigate the optimum temperature for the species, *Porites porites*. Coral fragments from different environments in the wild were collected and then subjected to different temperatures in the laboratory. The amount of stress was measured for each temperature. The optimal temperature is taken as the temperature that induced the least stress in the coral fragments.

The results showed that optimum temperature varies slightly within population and varies more between the two populations of *P. porites* found in the Northern South China Sea (NSCS) and Singapore Straits (SS).

Table 8.1

coral fragments from	optimum temperature (°C)												
	24.0	24.5	25.0	25.5	26.0	26.5	27.0	27.5	28.0	28.5	29.0	29.5	30.0
NSCS population	4	21	24	8									
SS population										3	15	26	7
F1				3	5	12	13	8	4				
F2		6	8	10	22	19	15	8	7	3	2		

Numbers shown in the table represents the number of coral fragments suited for the different temperature.

- (i) State the term used to describe the range of phenotypes shown in Table 8.1.

1. Continuous variation; [1]

- (ii) Explain the genetic basis behind the wide range of optimum temperatures.

- 1. Quantitative characteristic controlled by polygenes / more than 2 pairs of independent genes;**
- 2. Cumulative / additive effect on the trait without complete dominance;**
- 3. In F2, the segregation & recombination of these 3 allele pair produce the greatest variations;**
- 4. F1 fully heterozygous, intermediate;** [2]

- (iii) Coral fragments within each parental population are pure-breeding, yet the coral fragments within each parental population showed variation in optimum temperature. The variation in F1 is comparable to the average of the parental variations while the variation in F2 is greater than that found in the parental populations or the F1.

Explain why there is variation in each parental population, F1, and F2.

1. Parental lines are pure breeding, hence homozygotes at loci, variation due to environment;
2. F1 offspring should all be heterozygous at all loci and variation around the mean is derived from environmental influence;
3. F2 variation is called by individuals being genetic variation and environmental factors;

[3]

- (b) In another study, a sample of 50 coral larvae from two different lines of *P. porites* was infected with equal doses of bacterium (*Vibrio natriegens*). It was observed that different lines showed pronounced differences in mortality.

The remaining larvae from the two different lines were crossed to produce the F1 hybrid and infected with the same dose of *V. natriegens*.

Table 8.2 shows the results.

Table 8.2

line of <i>P. porites</i>	mortality / % of larvae infected
resistant	0
susceptible	100
cross between susceptible and resistant <i>P. porites</i> (F1 hybrid)	0

It is thought that resistance to *V. natriegens* is controlled by a single gene.

- (i) Using suitable symbols, draw a genetic diagram to show how resistance to *Vibrio* species could be inherited in the cross that produce the F1 hybrid shown in Table 8.2.

Perform monohybrid cross [3]

With parental genotype: RR x rr;

Punnet Square + correct circling of gametes;

F1 genotype: Rr;

- (ii) Calculate the probability of susceptible *P. porites* resulting from crosses between:

susceptible x F1 hybrid **50%;**

resistant x F1 hybrid **0%;** [2]

- (ii) Suggest **one** implication of this study to the environment.

Destruction of living population of coral reefs, may result in loss in diversity; [1]

[Total: 12]

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Name: _____

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Section E

For Examiner's Use	
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Section E

Answer the question in this section.

- 9 (a) The islands of Wan-an and Lanyu, separated by 25 km, support the two nesting populations of green turtle (*Chelonia mydas*).

Fig. 9.1 shows the position of the two islands around Taiwan and nesting beaches marked with rectangles.

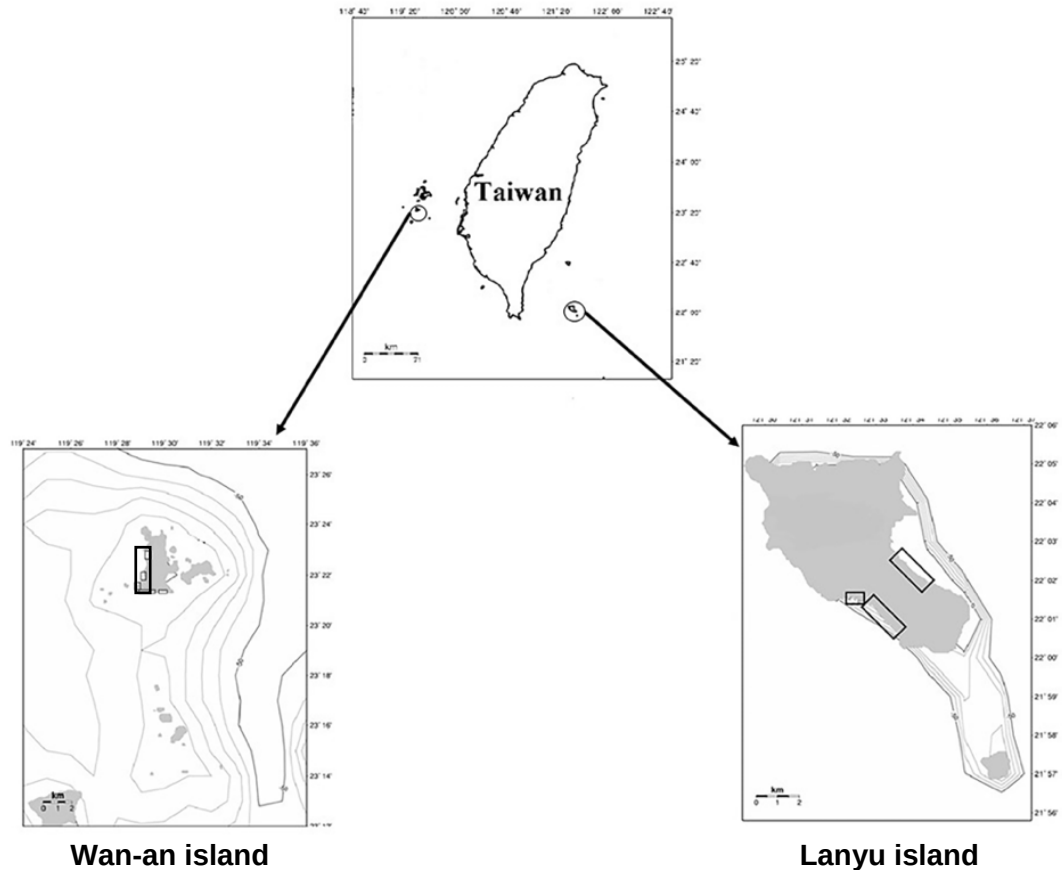


Fig. 9.1

A multi-parameter study was conducted on the two islands to observe the populations of green turtles from 1996 - 2016.

One parameter for the study is the number of new turtle nests in each island.

Fig. 9.2 shows the trend for Wan-an island.

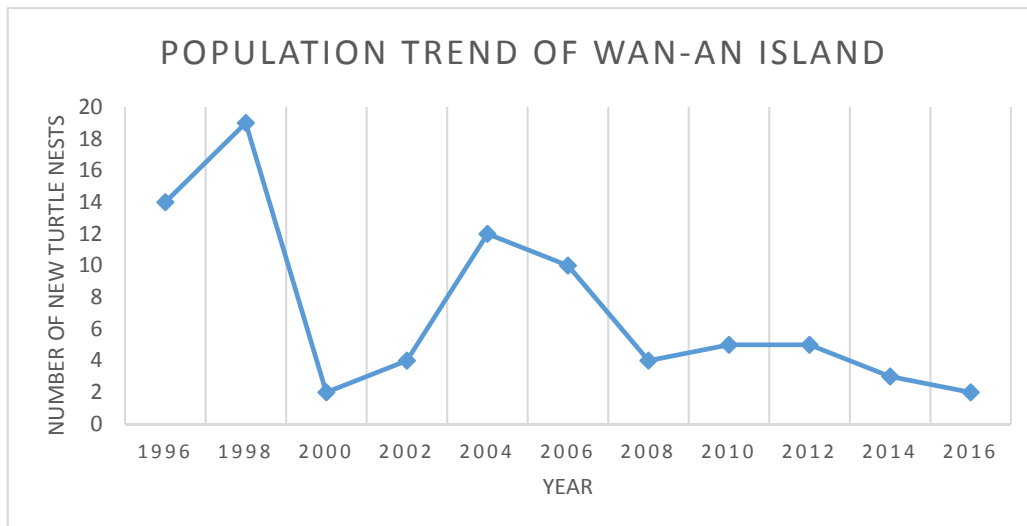


Fig. 9.2

(i) With reference to Fig. 9.2, describe the general trend observed.

- **Number of turtle nests decreased from 14 nests in 1996 to 2 nests in 2016;**

[1]

(ii) Suggest **one** reason for this trend.

- **Reclamation works on the island;**
- **Building of resorts on the island;**
- **Poachers on the islands;**
- **Decreased in number of mature turtles due to increase predation;**

[1]

- (b) The other parameters of the study include nest depth and number of eggs produced per season.

Table 9.3 shows the comparison of mean values for the two parameters between Lanyu (L) and Wan-an (W) islands from 1998 – 2014 and the t-test results.

Table 9.3

parameter	mean value	p-value
nest depth	W > L	< 0.001
number of eggs produced per season	W ~ L	0.583

- (i) With reference to Table 9.3, describe the results of the t-test.
- There is significant difference between the nest depth of Lanyu and Wan-An turtle populations;
 - There is no significant difference between the number of eggs produced per season of Lanyu and Wan-An Turtle population; [2]
- (ii) Based on your answer in (i), comment whether the results are sufficient to consider the two populations as separate species.
- Not sufficient;
 - Unable to tell whether they can still reproduce to produce fertile and viable offspring;
 - Ref: Biological speciation; [2]
- (iii) Explain how the data supports the theory of biological evolution as descent with modification.
- Genetic variations exist in green turtles of Taiwan;
 - Different selection pressure in Lanyu and Wan-an island;
 - Alleles coding for advantageous traits are selected for;
 - Higher survival and reproductive success;
 - Produce viable, fertile offspring;
 - Leading to changes in allele frequency;
 - Descent with modification; [4]

(iv) Explain why the population is the smallest unit that can evolve.

- Individual organisms cannot evolve;
- An individual has the same genome its entire life, except for mutations acquired during its lifetime;
- Natural selection acts on individuals in a population, but only population evolves;
- Evolution is a process that takes place over extended periods of time as alleles coding for adaptations are passed from one generation to another;

[4]

[Total: 14]